Yousef Khaled Gouda

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PROFESSIONAL SUMMARY

Senior Software Engineer and AI Systems Architect with 20+ years of experience building and deploying production AI systems at scale. Deep expertise in LLM fine-tuning and alignment, Arabic NLP, RAG systems, computer vision, and enterprise MLOps across GCP and AWS. Recognized pioneer in Arabic AI with production systems serving millions of daily requests; active contributor to AI evaluation, model quality, and open-source Arabic AI tooling.

EXPERIENCE

Quality Control Manager Sep 2026 – Present
RWS (TrainAI) *Remote*

- Lead quality-control reviews for software-engineering and AI-evaluation workstreams, auditing task design, implementation quality, evaluator consistency, and compliance with project standards.
- Review team deliverables across evaluation pipelines; identify recurring defects, provide actionable remediation guidance, and improve submission readiness.
- Monitor evaluator performance and consistency across concurrent workstreams; flag systematic quality issues and escalate to project leads.
- Develop and maintain internal QC checklists, review rubrics, and feedback templates to standardize evaluation processes.
- Coordinate across engineering, annotation, and project management teams to align quality standards with client requirements.

Senior Software Engineer / AI Evaluation Engineer (Contract) Jul 2026 – Nov 2026
Turing *Remote*

- Designed, reviewed, and remediated Harbor-structured AI evaluation tasks covering task.toml files, instructions, Docker environments, reference solutions, deterministic verifiers, automated tests, golden trajectories, model evaluations, and review.csv documentation.
- Diagnosed verifier-fairness, hidden-requirement, semantic-equivalence, malformed-input, CLI-safety, reproducibility, and container-configuration issues across PreQC, QC V2, Oracle, GLM, stability, and Final QC phases; delivered submission-ready packages.
- Validated cross-model behavior across multiple LLMs; ensured task integrity and consistent evaluation outcomes under varied model responses.
- Authored clear task instructions and reference solutions that met Harbor specification requirements; remediated flagged submissions through targeted debugging cycles.
- Collaborated with Turing review team to address edge cases in verifier logic and improve deterministic test coverage across task categories.

AI Code Quality and Evaluation Specialist (Claude Code Evaluation) Mar 2025 – Oct 2026
Alignerr *Remote*

- Evaluated Claude Code outputs across GitHub repositories and SWE-bench-style software-engineering tasks spanning multiple domains and programming languages.
- Reviewed AI-generated software implementations for correctness, test coverage, code quality, requirement adherence, and production readiness using structured evaluation criteria.
- Conducted GitHub repository analysis to assess code organization, documentation quality, and alignment with task specifications.
- Delivered structured technical feedback identifying failure modes in model-generated code; contributed to improving model reliability across coding benchmarks.
- Assessed edge-case handling, security considerations, and algorithmic efficiency in AI-generated C++ and Python solutions.
- Selected Project Assignments (12): Agent Error Analysis Specialist (Mar-Apr 2025); Pulsar Word Alignment Eval (May-Jun 2025); ML Evals (Jun-Jul 2025); OpenClaw Traces Licensing (Jul-Aug 2025); FCP Eval 2 (Aug-Sep 2025); Code Human Preference, Claude Code Part 2 (Sep-Oct 2025); FCP Financial Eval (Oct-Nov 2025); Terminator R&D/Upload/Eval (Nov-Dec 2025; Apr-May 2026; Jun-Jul 2026); Project Function Calling (Dec 2025-Jan 2026); Arabic Front Desk Coordinator (Jan-Feb 2026); Perfect WebGen V2 (Feb-Mar 2026); Arabic Front Desk Receptionist (Mar-Apr 2026).

AI Training Contributor (ChatGPT) 2025 – 2026
Handshake AI *Remote*

- Provided human feedback on model-generated responses across diverse task types to support ChatGPT model training.
- Evaluated instruction-following quality, factual accuracy, response completeness, and tone appropriateness using structured rating criteria.
- Performed comparative response evaluation and preference ranking to support RLHF training pipelines.
- Reviewed answer quality across coding, reasoning, and open-ended generation tasks; provided actionable written feedback to improve model performance.

Software Engineering Contributor (C++) 2024 – 2025

Outlier AI *Remote*

- Completed software-engineering tasks involving C++ through the Outlier AI platform, contributing to AI model training and evaluation for code generation.
- Evaluated AI-generated C++ solutions for correctness, algorithmic efficiency, code quality, and adherence to problem specifications.
- Developed and validated test cases to verify solution behavior across edge cases; identified logical errors and provided corrective implementations.
- Reviewed AI-generated code for memory safety, runtime correctness, and compliance with C++ best practices.

Senior Software Engineer and AI Systems Architect Jan 2022 – Present

Freelance / Remote (International Clients: UAE, Saudi Arabia, Europe) *Cairo, Egypt*

- Designed, architected, and developed production-grade software applications and enterprise systems, covering backend architecture, API development, database integration, and application deployment.
- Architected scalable backend microservices and RESTful APIs using Python (FastAPI) and C++, integrating business logic, databases, and AI services for enterprise applications.
- Designed and deployed cloud-native applications across GCP and AWS using Docker, Kubernetes, and automated CI/CD pipelines with GitHub Actions, focusing on scalability, reliability, and maintainability.
- Engineered data-processing systems and database architectures using SQL, PostgreSQL, MongoDB, and vector databases to support high-volume applications and intelligent information systems.
- Led Arabic LLM alignment projects using 80K instruction samples, QLoRA, and DPO.
- Designed production RAG systems using LangChain, Qdrant, and cross-encoder reranking over 10K+ Arabic legal documents, serving 3,000+ daily queries.
- Established MLOps standards using MLflow, Weights and Biases, Grafana, and Prometheus across client projects.

Lead AI Engineer and Research Manager Mar 2018 – Dec 2021

Egyptian Company for Software and AI Solutions (ECSA) *Cairo, Egypt*

- Led a team of 9 ML engineers building NLP and computer vision products for Arabic-language markets across Egypt, Saudi Arabia, and the UAE.
- Architected and productionized an Arabic text classification platform handling 10M+ documents/day using fine-tuned BERT multilingual, later migrated to AraBERT; accuracy improved from 81% to 93%.
- Built an Arabic speech recognition pipeline: Wav2Vec 2.0 fine-tuned on a 200-hour Egyptian dialect corpus; deployed as a REST API serving 50K+ daily transcriptions.
- Designed a real-time object detection and tracking system for a smart-city surveillance contract: YOLOv4 + DeepSORT processing 30 camera streams simultaneously on a 4x V100 GPU cluster.
- Introduced Transformer-based models to the team in 2018; led internal study group and migrated 3 production NLP pipelines to BERT within 6 months.
- Established the company's first MLOps infrastructure: Kubernetes-based model serving, A/B testing framework, automated retraining pipelines, and model drift monitoring.

Senior Machine Learning Engineer Jun 2014 – Feb 2018

Softex Egypt / Information Technology Industry Development Agency (ITIDA) Projects *Cairo, Egypt*

- Pioneered adoption of deep learning in Egypt's public-sector AI initiatives; among the first engineers in Egypt to deploy CNNs to production (2015) for document digitization and automated form recognition.
- Built an Arabic handwriting recognition system (CNN + CTC loss) for Egypt's national ID digitization project; processed 2M+ ID card images at 96.2% character accuracy.
- Developed license plate recognition for the Egyptian traffic authority: custom CNN trained on 50K Egyptian plates in Theano/Keras; 94% recognition accuracy in real traffic conditions.
- Led migration from Theano to TensorFlow (2015-2016) across 5 active projects; authored internal migration guide used by 30+ engineers.
- Built an RNN/LSTM-based Arabic text generation and autocomplete system; early Arabic NLP product deployed in a government document management platform.
- Mentored 6 junior engineers; introduced weekly deep learning paper reading sessions that became a company-wide practice.

Machine Learning Engineer, Mid-Senior Level Jan 2010 – May 2014

Raya Contact Center / Raya IT *Cairo and Alexandria, Egypt*

- Designed and deployed ML-powered predictive analytics systems for Egypt's largest contact center operator: churn prediction (XGBoost, 88% AUC), call volume forecasting (ARIMA + gradient boosting), and agent performance scoring.
- Built a real-time fraud detection pipeline using anomaly detection (Isolation Forest, One-Class SVM) on call records; reduced fraudulent call losses by 34% in the first year.
- Led data science team of 4 engineers; introduced reproducible ML workflows with versioned datasets, model registries, and automated evaluation reports.
- Worked extensively with Python, R, Pandas, NumPy, Scikit-learn, and SQL on datasets ranging from 10M to 500M rows.

- Delivered Arabic sentiment analysis v1 using SVM + TF-IDF on customer feedback transcripts; company's first NLP product (2012).

Software Engineer, then Data Analyst Jul 2005 – Dec 2009

Xceed Contact Center Egypt / Link Development *Cairo, Egypt*

- Built C++ and Python data processing tools for large-scale call-center analytics platforms; developed statistical reporting systems and early predictive models using logistic regression and decision trees.
- Implemented automated data pipelines processing 5M+ daily records; reduced manual reporting effort by 70%.
- Led SQL optimization efforts reducing query execution time by 60% across reporting systems; earned Oracle Database and Microsoft certifications.

PROJECTS

Multi-Agent AI Research and Report Generation System 2024 – 2024

- Autonomous research agent that searches ArXiv and the web, parses papers, synthesizes cross-source insights, and generates structured technical reports in Arabic and English.
- Stateful LangGraph orchestration with tool use, episodic memory, and human-in-the-loop approval for high-stakes actions; deployed as a full-stack web app.
- Stack: LangGraph, Claude API, ArXiv API, SerpAPI, FastAPI, React, Mem0

Enterprise Arabic LLM Platform — Fine-Tuning and Deployment 2023 – 2024

- End-to-end Arabic LLM: curated 80K instruction samples, QLoRA fine-tuning on 8xA100, DPO alignment with 1K preference pairs; deployed via vLLM on GCP with auto-scaling.
- Production system serving 50K+ daily Arabic queries; P95 latency 420ms; 82% human preference score in blind evaluation vs. GPT-3.5-turbo at 71%.
- Full MLOps stack: MLflow registry, W&B training dashboard, Langfuse observability, Prometheus/Grafana production monitoring.
- Stack: LLaMA 3 8B/70B, QLoRA, Axolotl, DPO, TRL, vLLM, GCP, Qdrant, LangChain

Production RAG — Arabic Legal Document Intelligence 2023 – 2023

- RAG platform over 10K+ Arabic legal documents; hybrid BM25 + dense retrieval + cross-encoder reranking; 88% accuracy on 500-query gold standard.
- Containerized microservice architecture on GCP Cloud Run; 3,000+ daily queries at P95 under 900ms latency.
- Stack: LangChain, Qdrant, Multilingual-E5, Cross-encoder Reranking, FastAPI, Docker

Arabic Text Classification Platform — 10M Documents/Day 2018 – 2019

- Production NLP platform classifying Arabic news, social media, and government documents into 12 categories at 10M+ documents/day throughput.
- Migrated from SVM+TF-IDF (81% accuracy) to AraBERT (93%); Kubernetes autoscaling; Redis response caching; zero-downtime rolling deployments.
- Stack: AraBERT, TensorFlow Serving, Kubernetes, Redis, PostgreSQL, REST API

Smart-City Surveillance — Real-Time Multi-Camera Tracking 2019 – 2020

- Multi-object detection and tracking across 30 simultaneous RTSP camera streams; YOLOv4 + TensorRT INT8 at 28 FPS per stream on a 4xV100 cluster.
- Real-time alert system for crowd density, loitering, and perimeter breach; integrated with city operations dashboard via WebSocket streaming.
- Stack: YOLOv4, DeepSORT, OpenCV, TensorRT, NVIDIA V100 Cluster, RTSP Streams

Arabic Speech Recognition API — Egyptian Dialect 2020 – 2021

- Fine-tuned Wav2Vec 2.0 on 200-hour Egyptian Arabic speech corpus (call center recordings); 19% WER, best-in-class for Egyptian dialect at the time.
- REST API serving 50K+ daily transcriptions; async processing queue with Redis; auto-scaling Docker deployment on AWS ECS.
- Stack: Wav2Vec 2.0, Hugging Face, 200h Egyptian Corpus, FastAPI, Redis, Docker

National ID Digitization — Arabic OCR Pipeline 2015 – 2016

- CNN-based Arabic handwriting OCR for Egypt's national ID digitization program; 96.2% character accuracy on 2M+ scanned ID cards.
- One of Egypt's first production CNN deployments; custom data augmentation pipeline for handling handwriting variability and scan quality issues.
- Stack: Keras (Theano backend), CNN, CTC Loss, 2M+ ID Images, Python, PostgreSQL

Contact Center Fraud Detection and Churn Prediction System 2010 – 2013

- Real-time fraud detection using Isolation Forest + One-Class SVM on call records; reduced fraudulent call losses by 34% in year one.
- Churn prediction model (XGBoost, AUC = 0.88) deployed to guide retention campaigns; directly attributed to 12% improvement in customer retention rate.

- Call volume forecasting (ARIMA + gradient boosting) improved staffing efficiency by 23%; one of Egypt's first ML-driven contact center optimization systems.
- Stack: Python, Scikit-learn, XGBoost, ARIMA, SQL, Hadoop (early cluster)

SKILLS

LLMs and Generative AI: LLaMA 2/3, Mistral, Mixtral MoE, GPT-4o, Claude 3.5, Gemini, Fine-tuning (LoRA, QLoRA, Full SFT), RLHF / DPO / GRPO, RAG Systems, LangChain, LlamaIndex, LangGraph, DSPy, Hugging Face PEFT/TRL, Axolotl, vLLM, TensorRT
AI Evaluation and Quality Control: LLM Response Evaluation, Automated Verifier Design, Task Benchmarking, SWE-bench Code Evaluation, Claude Code Evaluation, Harbor Evaluation Framework, QC Workflow Auditing
Machine Learning and Computer Vision: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, AraBERT, OpenCV, YOLOv4/v5/v8, DeepSORT, Arabic NLP and OCR, Speech Recognition (Wav2Vec 2.0)
Software Engineering and Architecture: System Architecture, Distributed Systems, Microservices, RESTful API Design, Object-Oriented Programming (OOP), Design Patterns, Data Structures and Algorithms, Memory Management, Performance Optimization
Programming Languages: Python (Expert, 20+ years), C++ (Advanced, 18+ years), C (Advanced), SQL (Expert), Bash / Linux CLI, JavaScript / TypeScript, R, CUDA
DevOps, Cloud and Infrastructure: Docker, Kubernetes, GCP (Cloud Run, Vertex AI), AWS (ECS, SageMaker), GitHub Actions CI/CD, Terraform, Prometheus, Grafana, Redis, Nginx, Linux System Administration
Backend and Database Engineering: FastAPI, Django, Flask, Celery, PostgreSQL, MongoDB, Elasticsearch, Qdrant, Weaviate, FAISS, Pinecone, Database Optimization, Asynchronous Processing
Frontend and Full-Stack Development: React, JavaScript, TypeScript, Node.js, HTML5, CSS3, REST API Integration
Languages: Arabic (Native: Egyptian and Modern Standard Arabic), English (Fluent, near-native)

CERTIFICATIONS

- TensorFlow Developer Professional Certificate — Google / Coursera (2020)
- CCNA: Cisco Certified Network Associate — Cisco (2008)
- Linux Professional Institute Certification (LPIC-1) — LPI (2007)
- Microsoft Certified: MCP — Programming in C# .NET (2007)
- Oracle Database 10g: SQL Fundamentals and PL/SQL — Oracle Certified Associate (2006)
- Multimodal RAG: Chat with Videos — DeepLearning.AI (2025)
- JavaScript AI SDK and Vercel AI — Vercel / DeepLearning.AI (2025)
- AI Agents in LangGraph — DeepLearning.AI + LangChain (2024)
- Multi AI Agent Systems with crewAI — DeepLearning.AI (2024)
- Building Agentic RAG with LlamaIndex — DeepLearning.AI (2024)
- Efficiently Serving LLMs — DeepLearning.AI (2024)
- Mechanistic Interpretability — ARENA / Neel Nanda (2024)
- AI Safety Fundamentals: Alignment Course — BlueDot Impact (2024)
- RL from Human Feedback (RLHF) — DeepLearning.AI (2023)
- Building and Evaluating Advanced RAG — DeepLearning.AI + LlamaIndex (2023)
- Finetuning Large Language Models — DeepLearning.AI (2023)
- LangChain for LLM Application Development — DeepLearning.AI (2023)
- Building Systems with the ChatGPT API — DeepLearning.AI (2023)
- MLOps Specialization (4 courses) — DeepLearning.AI / Coursera (2022)
- Full Stack Deep Learning — FSDL (2022)
- Hugging Face NLP Course — Hugging Face (2022)
- Hugging Face Diffusion Models Course — Hugging Face (2022)
- Practical Deep Learning for Coders Part 2 — fast.ai (2022)
- Computer Vision Specialization — DeepLearning.AI / Coursera (2021)
- ML Pipelines on Google Cloud — Google Cloud Training (2021)
- Transformer Models and BERT Model — Google Cloud / Coursera (2020)
- Applied AI with DeepLearning — IBM / Coursera (2020)
- Natural Language Processing Specialization (4 courses) — DeepLearning.AI / Coursera (2019)
- Practical Deep Learning for Coders Part 1 — fast.ai / Jeremy Howard (2019)
- PyTorch Scholarship Challenge — Facebook AI / Udacity (2018)
- Advanced Machine Learning Specialization — HSE Moscow / Coursera (2018)
- Sequence Models — Andrew Ng / DeepLearning.AI / Coursera (2018)
- Deep Learning Specialization (5 courses) — Andrew Ng / DeepLearning.AI (2017)

- Data Science Specialization (10 courses) — Johns Hopkins / Coursera (2014)
- Neural Networks for Machine Learning — Geoffrey Hinton / Coursera (2013)
- Mining Massive Datasets — Stanford / Coursera (2013)
- Probabilistic Graphical Models — Daphne Koller / Stanford / Coursera (2013)
- Computing for Data Analysis (R) — Roger Peng / Johns Hopkins / Coursera (2013)
- Statistical Learning — Stanford Online / Trevor Hastie and Robert Tibshirani (2013)
- Machine Learning — Andrew Ng / Stanford / Coursera (2012, original course)
- Algorithms: Design and Analysis — Tim Roughgarden / Stanford / Coursera (2012)
- Audited: CS224N: NLP with Deep Learning — Stanford Online (2019)
- Audited: CS285: Deep Reinforcement Learning — UC Berkeley (2018)
- Audited: CS231n: CNNs for Visual Recognition — Stanford Online (2016)
- Project Management Fundamentals — PMI / Cairo Chapter (2009)

Selected Research And Technical Publications

- All technical reports published on GitHub (github.com/yousefkhaled2005) and Hugging Face Hub. Selected reports presented at Egyptian and regional AI meetups.
- Arabic LLM Evaluation Framework — Hallucination and Factuality Benchmark (2025): Comprehensive evaluation of 6 frontier models on Arabic factual QA (legal, medical, historical domains); custom 3-tier hallucination taxonomy. Showed 28% higher hallucination rate in Arabic vs. English for same models; results submitted to Arabic NLP Workshop proceedings. Stack: GPT-4o, LLaMA 3, Mistral, Gemini, Claude, 1,200-query Arabic test suite.
- RLHF Preference Data Quality — Impact on Arabic LLM Alignment (2024): Empirical study on how annotation agreement thresholds affect downstream DPO alignment; showed Kappa >= 0.78 is required for consistent alignment gains. Practical annotation guidelines adopted by 2 Gulf-based AI teams as their internal standard. Stack: DPO, TRL, 1,000-pair annotated dataset, Kappa agreement analysis.
- RAG vs. Fine-Tuning for Arabic Domain QA — Comparative Benchmark (2023): Systematic comparison across 4 domains; RAG superior for factual recall (88% vs 84%), fine-tuning superior for latency (3x faster); hybrid architecture proposed. Stack: LangChain, Qdrant, QLoRA, Arabic Legal and Medical Corpora, 500-query eval.
- Efficient Arabic LLM Serving — Inference Engine Benchmark (2023): Head-to-head benchmark of vLLM, llama.cpp, and Triton on LLaMA 2/3 variants; comprehensive throughput, latency, and memory analysis across 6 hardware configs. Stack: vLLM, llama.cpp, GGUF INT4/INT8, Triton, T4/A100 GPU, CPU benchmarks.
- AraBERT vs. Multilingual BERT — Systematic Benchmark for Arabic NLP (2020): First systematic Arabic BERT comparison covering sentiment, NER, QA, and dialect ID; AraBERT-v2 outperforms mBERT by 4-11 F1 points across tasks. Widely cited in Egyptian AI community; report used as baseline by 3 university NLP research groups.
- YOLOv4 vs. EfficientDet for Arabic OCR Scene Text Detection (2020): Compared detection architectures on Arabic scene text; YOLOv4 achieved mAP@0.5 = 0.79 vs. EfficientDet-D3 at 0.77 with 2x faster inference.
- Arabic Dialect Identification at Scale — Dataset and Baseline Study (2019): Created and released a 5,000-sentence Arabic dialect evaluation dataset (Egyptian, Gulf, Levantine, Maghrebi, MSA). Baseline models and annotation guide published on GitHub; dataset downloaded 800+ times on Hugging Face Hub.
- Deep Neural Networks for Egyptian Arabic ASR — Early Study (2017): Early experimental study of deep learning for Egyptian Arabic speech recognition; LSTM+CTC achieved 34% WER, an 18% improvement over the GMM-HMM baseline.
- Ensemble Methods for Arabic Sentiment Analysis — Pre-Transformer Era (2015): Systematic comparison of 8 classical ML models on Arabic sentiment; SVM + TF-IDF bigrams achieved 86.4% accuracy. Published as internal technical report at Softex Egypt.

Open Source And Github Portfolio

- github.com/yousefkhaled2005 | huggingface.co/yousef-khaled: 20+ public repositories, 8 Hugging Face models, 3 public datasets.
- Models on Hugging Face Hub: yousef-khaled/arabert-sentiment-v2 (F1=0.914); yousef-khaled/arabic-dialect-classifier (5-class dialect ID, F1=0.87); yousef-khaled/llama3-arabic-instruct (QLoRA fine-tuned Arabic LLM); yousef-khaled/arabic-hallucination-detector; yousef-khaled/arabic-ner-v2 (NER: PER/ORG/LOC/DATE); yousef-khaled/wav2vec2-egyptian-arabic (ASR); yousef-khaled/arabic-dialect-eval-1k; yousef-khaled/arabic-rlhf-preferences (1K-pair preference dataset).
- Key GitHub repositories: arabic-llm-platform (QLoRA+DPO fine-tuning pipeline, Axolotl); production-rag-arabic (LangChain+Qdrant RAG for Arabic legal docs); multi-agent-research (LangGraph autonomous research agent); arabic-asr-pipeline (Wav2Vec 2.0 Egyptian dialect ASR); llm-inference-benchmark (vLLM vs llama.cpp vs Triton); arabic-bert-benchmark (AraBERT vs mBERT); arabic-dialect-dataset; yolov4-arabic-ocr; rlhf-annotation-tools; ml-system-design.

Career Highlights

- 20+ Years of End-to-End AI Engineering: From classical ML (2005) to deep learning (2015) to LLMs and Agentic AI (2024-2026).

- Contributed to AI model training, evaluation, and quality control across international platforms, including projects involving leading large language models.
- Arabic AI Leadership: Pioneer in Arabic NLP, ASR, and Arabic LLM fine-tuning; built production systems serving millions of users across Egypt, UAE, and Saudi Arabia.
- Scale and Production Reliability: Successfully architected platforms handling up to 10M+ documents/day and real-time multi-camera vision systems across multi-GPU clusters.
- Open-Source and Community Impact: 20+ public repos, 8 Hugging Face models, and 15+ technical research reports shaping Arabic AI evaluation benchmarks.

CONTINUOUS LEARNING

- Pattern Recognition and Machine Learning — Christopher Bishop (self-study, 2009-2011).
- The Elements of Statistical Learning — Hastie, Tibshirani, and Friedman (self-study, 2010).
- R Programming — Self-study / Coursera early access (2011).

EDUCATION

Germany
Ph.D. in Computer Science. *2010*

Cairo University, Faculty of Engineering Cairo, Egypt
M.Sc. in Computer Science. *2006*

Benha University, Faculty of Engineering Banha, Egypt
B.Sc. in Computer and Control Systems Engineering. Graduated with Honours, Grade: Excellent. Thesis: Adaptive Control Systems for Industrial Process Automation Using Fuzzy Logic and Neural Networks. Final-year project: PID auto-tuning system using a neural network controller, implemented in C and tested on AVR hardware. *2004*